

1. General Description

The EMS3518 is high-performance dual single-pole single-throw (SPST) depletion switches. Depletion technology enables the device to conduct signals in the absence of V_{DDx} and to isolate signals in the presence of V_{DDx} . The EMS3518 can pass or isolate negative signal fluctuations as low as -3.0 V and is 5.0 V tolerable. The EMS3518's depletion gate control enables it to achieve exceptional THD+N performance while using very little power during signal conduction.

2. Features and Benefits

- Dual SPST depletion switches
- Normally closed when $V_{DDx} < 0.4$ V
- V_{SW} : - 3.0 V to 5.0 V
- R_{ON} : 0.85 Ω (typ.)
- $R_{FLAT(ON)}$: 8.7m Ω (typ.)
- THD+N: - 102dB

3. Applications

- Smart phones, Tablets, laptops
- Bluetooth headsets, Mobile accessories

4. Ordering Information

Table 1. Ordering information

Type number	Package		
	Name	Description	Quantity
EMS3518DRB	DFN 3 x 3 - 8L	DFN package, 8 leads, 3 mm × 3 mm; 0.65 mm pitch, 0.8 mm (Max) height	5000

5. Function Diagram

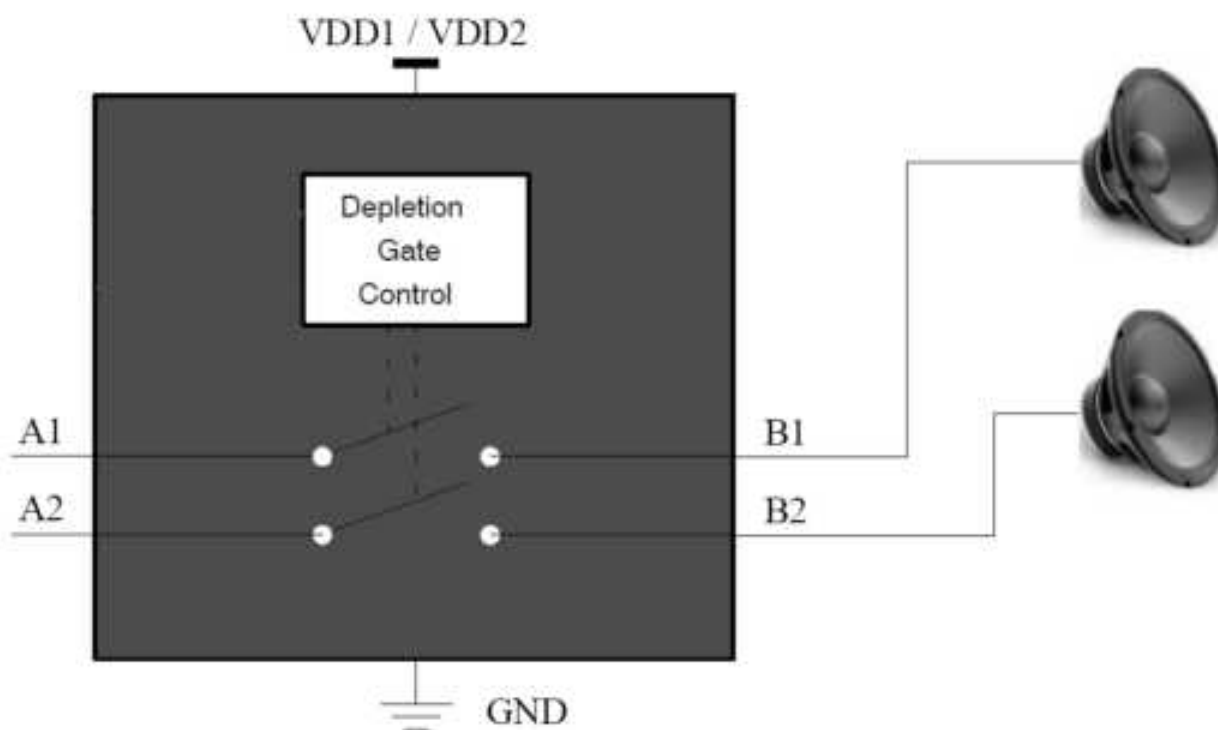


Fig 1. Function Diagram

6. Pinning Information

6.1. Pin map

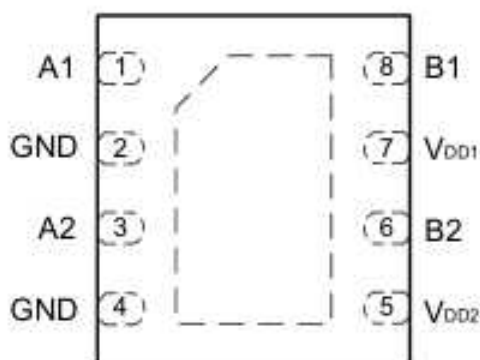


Fig 2. Top view pin configuration

6.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
A1	1	A-port of Switch One
GND	2,4	Ground
A2	3	A-port of Switch Two
V _{DD2}	5	Supply voltage (Switch Two is turned-on when V _{DD2} is low) of Switch Two
B2	6	B-port of Switch Two
V _{DD1}	7	Supply voltage (Switch One is turned-on when V _{DD1} is low) of Switch One
B1	8	B-port of Switch One

7. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

VDDx	Switch state
L	ON (Conducting)
H	OFF (Isolating)

8. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{DDx}	Supply voltage V_{DD1} or V_{DD2}		-0.5	5.5	V
V_{SW}	Switch voltage range	DC switch I/O voltage (switch conducting)	-3.5	5.5	V
		DC switch I/O voltage (switch isolated)	-3.5	5.5	V
I_{SW}	Maximum peak switch I/O current-pulsed at 1ms duration, <10% duty cycle			500	mA
T_{stg}	Storage Temperature		-65	150	°C
ESD	HBM	Human-body model	-5000	+5000	V
	CDM	Charged-device model	-2000	+2000	V
Latch-up				250	mA

9. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Max	Unit
V_{DDx}	Supply voltage V_{DD1} or V_{DD2}	Isolating	2.0	5.0	V
$V_{SW(OFF)}$	Switch off voltage	Isolating ($V_{DDx} = 2.0$ V)	-2.0	+2.0	V
		Isolating ($V_{DDx} = 2.5$ V)	-2.5	+2.5	V
		Isolating ($V_{DDx} = 3.3$ V)	-3.0	+3.3	V
		Isolating ($V_{DDx} = 4.0$ V)	-3.0	+4.0	V
		Isolating ($V_{DDx} = 5.0$ V)	-3.0	+5.0	V
$V_{SW(ON)}$	Switch on voltage	Conducting	-3.0	+5.0	V
T_A	Ambient Temperature		-40	85	°C

10. Electrical Characteristics

Table 6. Static characteristics

Unless otherwise specified, typical values are at $V_{DDx} = 0V$, $T_A = +25\text{ }^{\circ}C$.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
DC Electrical Characteristics						
I_{OFFX}	Switch off leakage current	$V_{DDx} = 3.3 \text{ to } 5.0 \text{ V}$, $Bx = GND$	$Ax = V_{DDx}$	0.2		μA
			$Ax = -1.5 \text{ V to } 1.5 \text{ V}$	0.7		μA
		$V_{DDx} = 2.0 \text{ to } 3.3 \text{ V}$, $Bx = GND$	$Ax = -1.5 \text{ V to } 1.5 \text{ V}$	0.5		μA
R_{ON}	Switch on resistance	$I_{SW} = 50 \text{ mA}$, $V_{SW} = -3 \text{ V to } 3 \text{ V}$		0.85		Ω
$R_{FLAT(ON)}$	On resistance flatness	$I_{SW} = 50 \text{ mA}$, $V_{SW} = -3 \text{ V to } 3 \text{ V}$		8.7		m Ω
C_{ON}	On capacitance	$V_{SW} = 0 \text{ dBm}$, $f = 1 \text{ MHz}$		30		pF
C_{OFF}	Off capacitance	$V_{SW} = 0 \text{ dBm}$, $f = 1 \text{ MHz}$		23		pF
AC Electronics Characteristics						
THD+N	Total harmonic distortion plus noise	$V_{SW} = 0.7 V_{RMS}$, $f = 1 \text{ kHz}$ @ $R_L = 32 \Omega$		-102		dB
OIRR	Off isolation rejection ratio	$V_{DDx} = 2 \text{ V}$, $V_{SW} < 1.7 V_{RMS}$, $f = 20 \text{ kHz}$	$R_L = 32 \Omega$	-115		dB
			$R_L = 1 \text{ k}\Omega$	-100		dB
		$V_{DDx} = 2.5 \text{ V}$, $V_{SW} < 1.7 V_{RMS}$, $f = 20 \text{ kHz}$	$R_L = 32 \Omega$	-115		dB
			$R_L = 1 \text{ k}\Omega$	-100		dB
		$V_{DDx} = 3.3 \text{ V}$, $V_{SW} < 2.1 V_{RMS}$, $f = 20 \text{ kHz}$	$R_L = 32 \Omega$	-115		dB
			$R_L = 1 \text{ k}\Omega$	-100		dB
		$V_{DDx} = 4 \text{ V}$, $V_{SW} < 2.1 V_{RMS}$, $f = 20 \text{ kHz}$	$R_L = 32 \Omega$	-115		dB
			$R_L = 1 \text{ k}\Omega$	-108		dB
		$V_{DDx} = 5 \text{ V}$, $V_{SW} < 2.1 V_{RMS}$, $f = 20 \text{ kHz}$	$R_L = 32 \Omega$	-115		dB
			$R_L = 1 \text{ k}\Omega$	-108		dB

EMS3518

Dual 5V SPST Depletion Audio Switches with Negative Swing

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
BW	Bandwidth	$R_L = 50\ \Omega$, $C_L = 0\ \text{pF}$		80		MHz
Supply currents						
I_{DDX}	Quiescent current	$V_{DDX} = 2.0\ \text{V}$		30		μA
		$V_{DDX} = 2.5\ \text{V}$		37		μA
		$V_{DDX} = 3.3\ \text{V}$		46		μA
		$V_{DDX} = 5.0\ \text{V}$		79		μA
I_{DISX}	Disable current	$V_{DDX} \leq 0.2\ \text{V}$		0.1		μA
Control logic						
R_{PD}	V_{DDX} pull-down resistance	$V_{DDX} \leq 0.2\ \text{V}$		5.5		M Ω
V_{DOH}	V_{DDX} high voltage		2.0			V
V_{DOL}	V_{DDX} low voltage				0.4	V
Timing characteristics						
t_{OFF}	Switch turn-off Time	$R_L = 1\ \text{k}\Omega$, $C_L = 10\ \text{pF}$, $V_{DDX} = 0\ \text{V}$ to $2\ \text{V}$, $V_{SW} = 1.5\ \text{V}$		170		μs
t_{ON}	Switch turn-on Time	$R_L = 1\ \text{k}\Omega$, $C_L = 10\ \text{pF}$, $V_{DDX} = 2\ \text{V}$ to $0\ \text{V}$, $V_{SW} = 1.5\ \text{V}$		200		μs
Oscillator frequency						
f_{OSC}	On-chip oscillator frequency	For reference only		3.3		MHz

11. Typical Characteristics

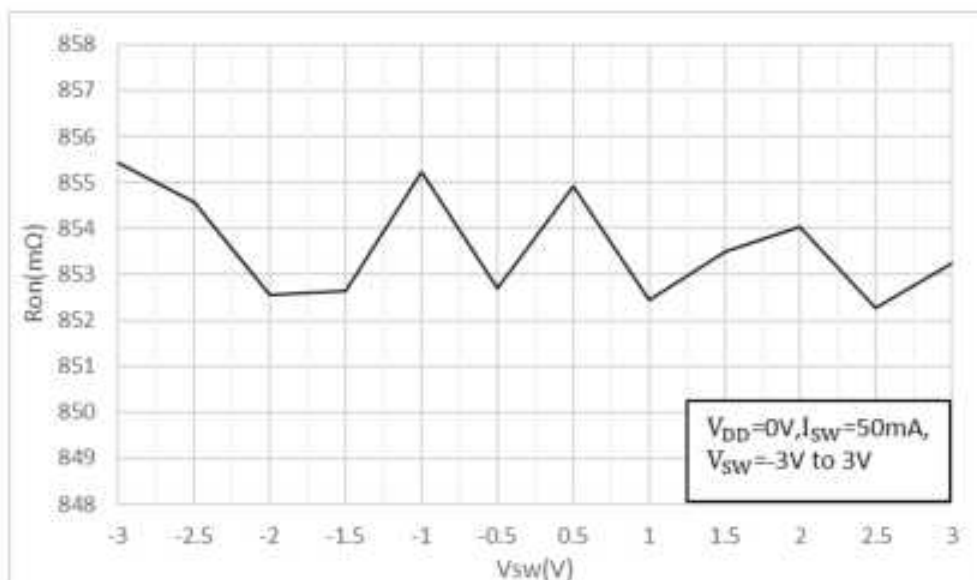


Fig 3. R_{ON} vs. V_{sw}

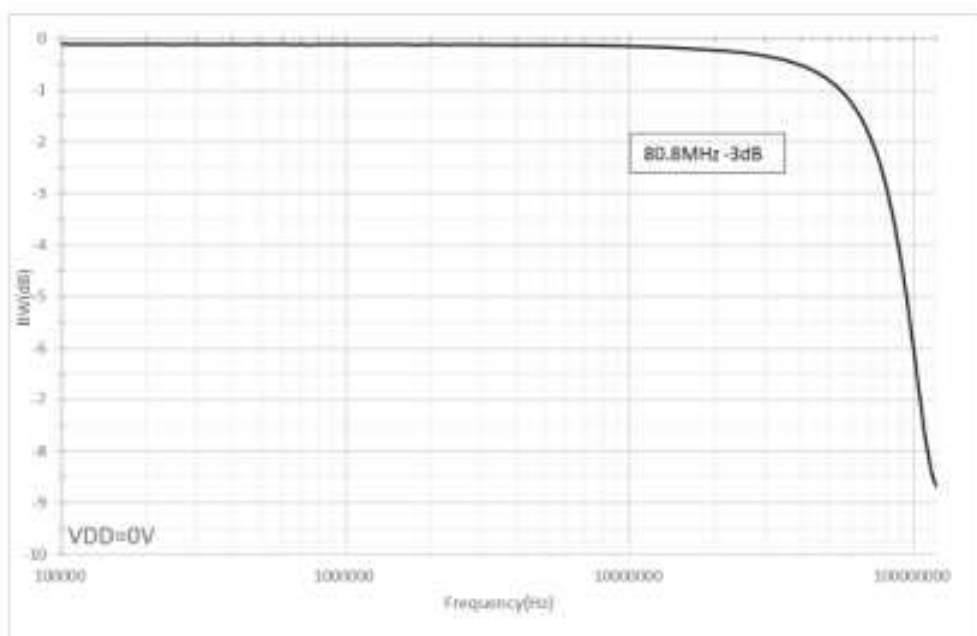


Fig 4. BW vs. Frequency

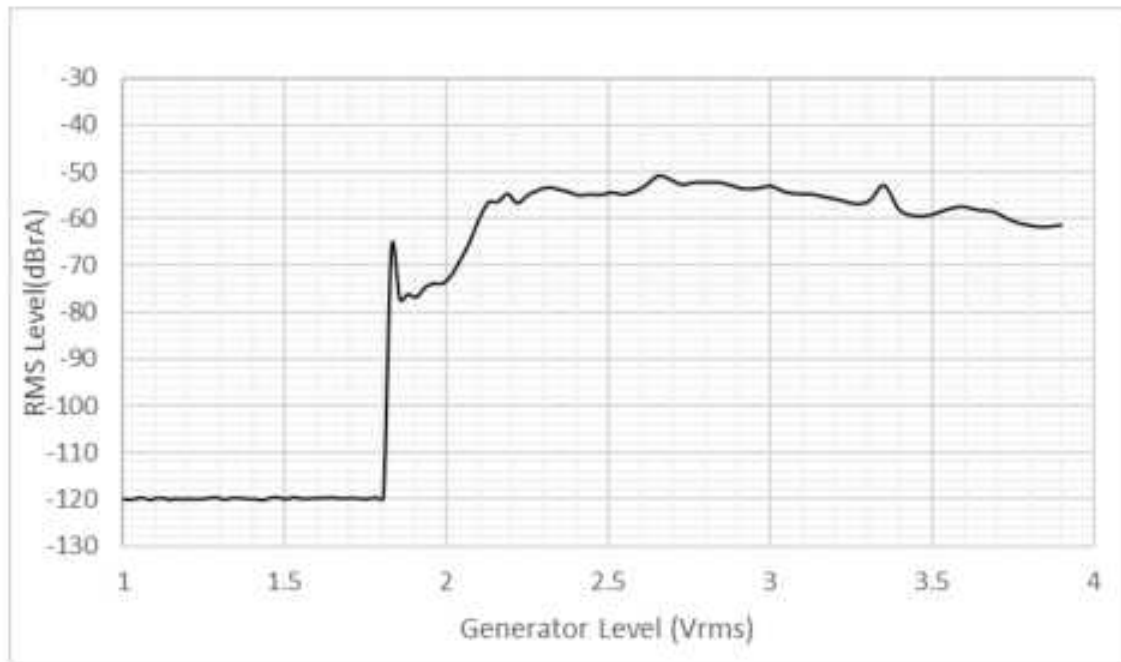


Fig 5. THD + N vs. Input with 33Ω Load

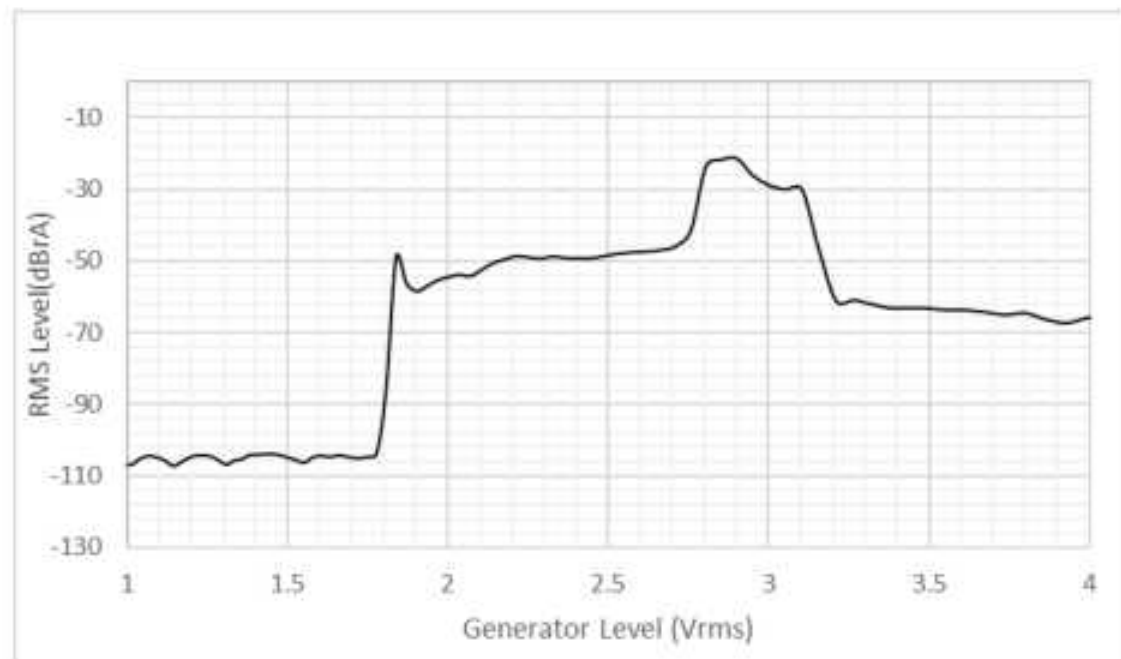


Fig 6. THD + N vs. Input with 1kΩ Load

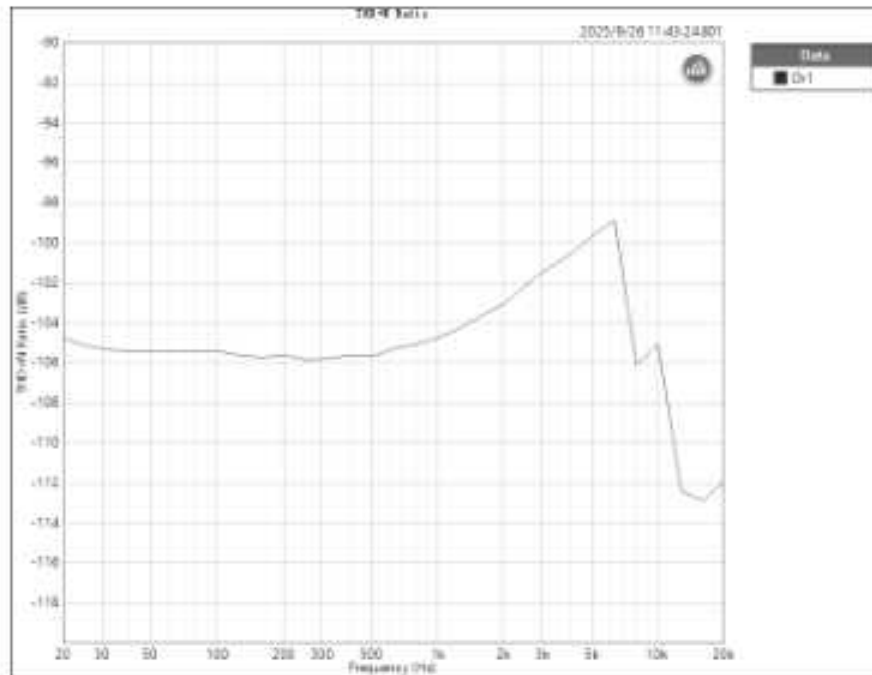


Fig 7. THD + N vs. Frequency with 33Ω Load

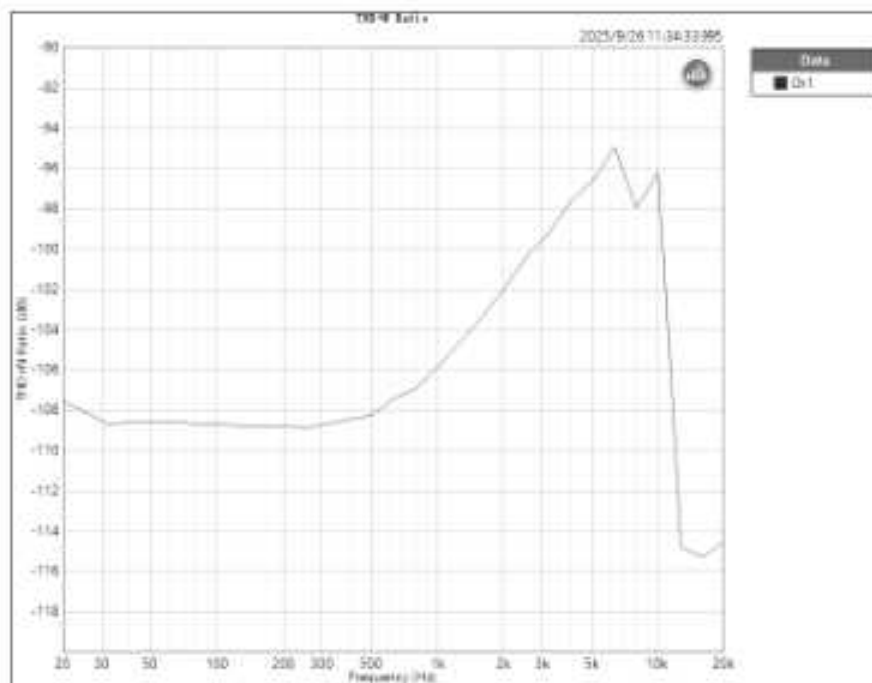
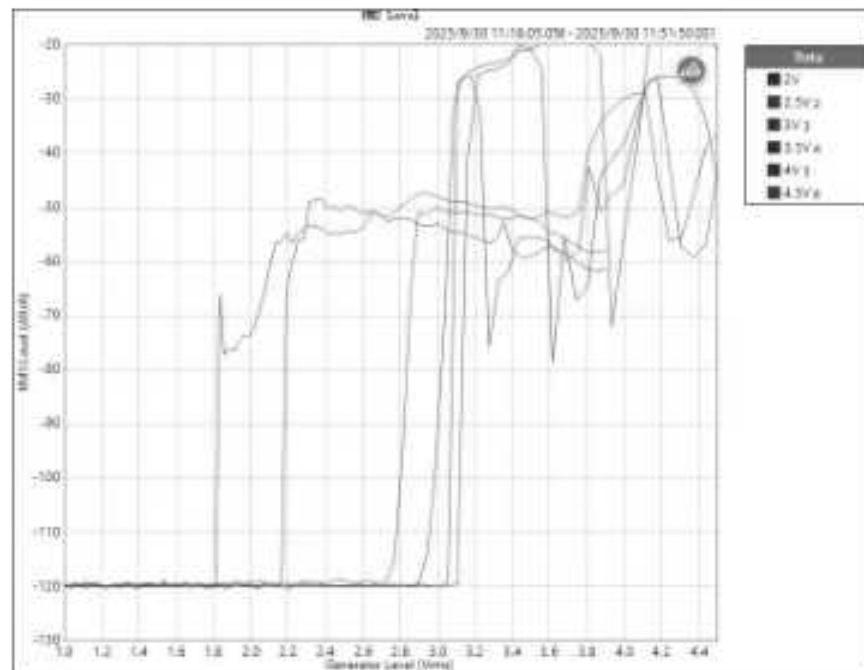
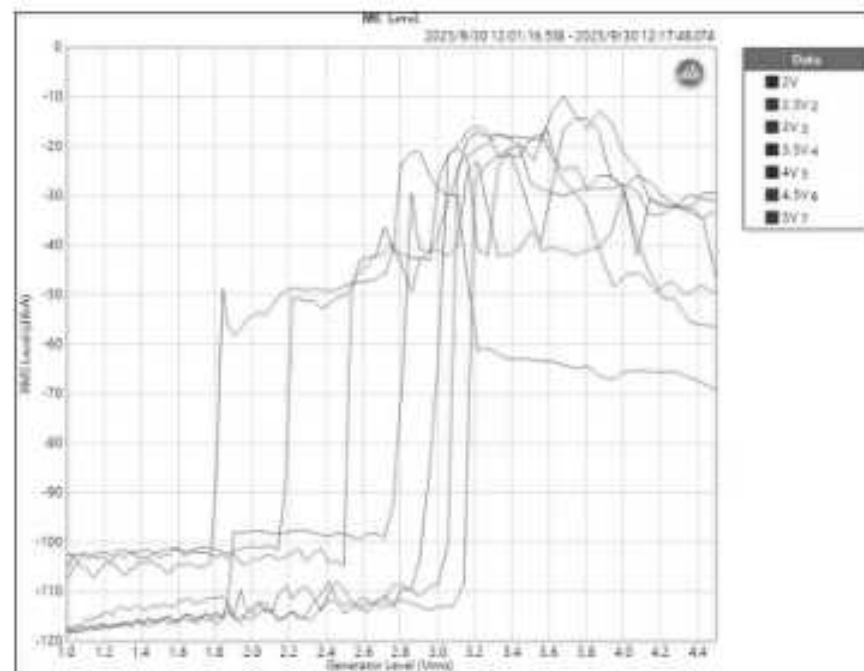
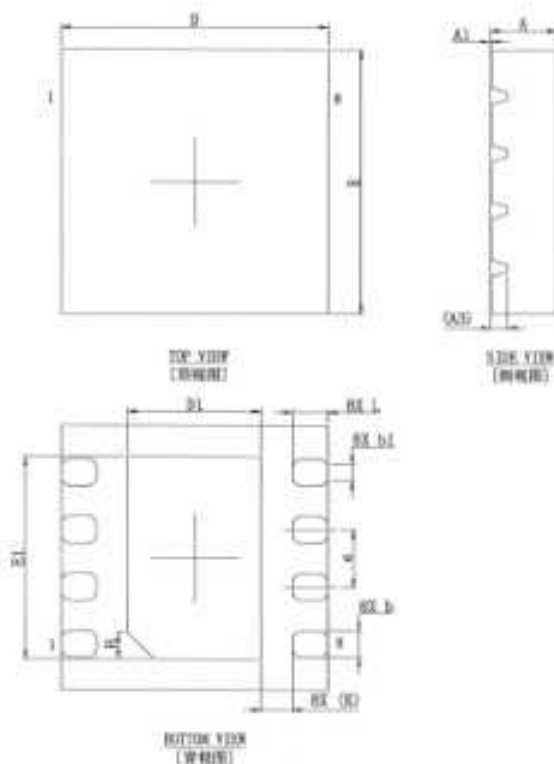


Fig 8. THD + N vs. Frequency with 1kΩ Load


Fig 9. Off-isolation vs. Input with 32Ω Load

Fig 10. Off-isolation vs. Input with 1kΩ Load

12. Package Outline

DFN 3x3 -8L



SYMBOL	MIN	NOM	MAX
A	0.700	0.750	0.800
A1	0.000	0.020	0.050
A3	0.203 REF		
b	0.250	0.300	0.350
b1	0.180 REF		
D	2.900	3.000	3.100
E	2.900	3.000	3.100
e	0.650 RSC		
D1	1.400	1.500	1.600
E1	2.200	2.300	2.400
L	0.300	0.400	0.500
K	0.350 REF		
H	0.300 REF		

13. Revision History

Table 7. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS3518 Rev.1.0	Oct 16, 2025	Product datasheet		